Fundamentals of Networking

- NETW191 Final Course Project
- **May 2024**
- **Barbara Espericueta**
- Prof. **Farooq Afzal**

Introduction

Fundamentals of Information Technology and Networking

Overall project: This course project covers the fundamentals of networking that play a vital role in the IoT world. The design and development process of building a network. This project includes SOHO router configurations, subnetting, connectivity testing, networking documentation, and SOHO wireless network security. In this project, we learn to emerge technology to build a network.

Objective: Learning of emerging technologies to build a network, including IoT.

Tools: aci learning, INFOSEC learning INC, Microsoft Visio, Google, live DeVry Study.

NETW191 Course Project

Module 2

IPv4 Addressing

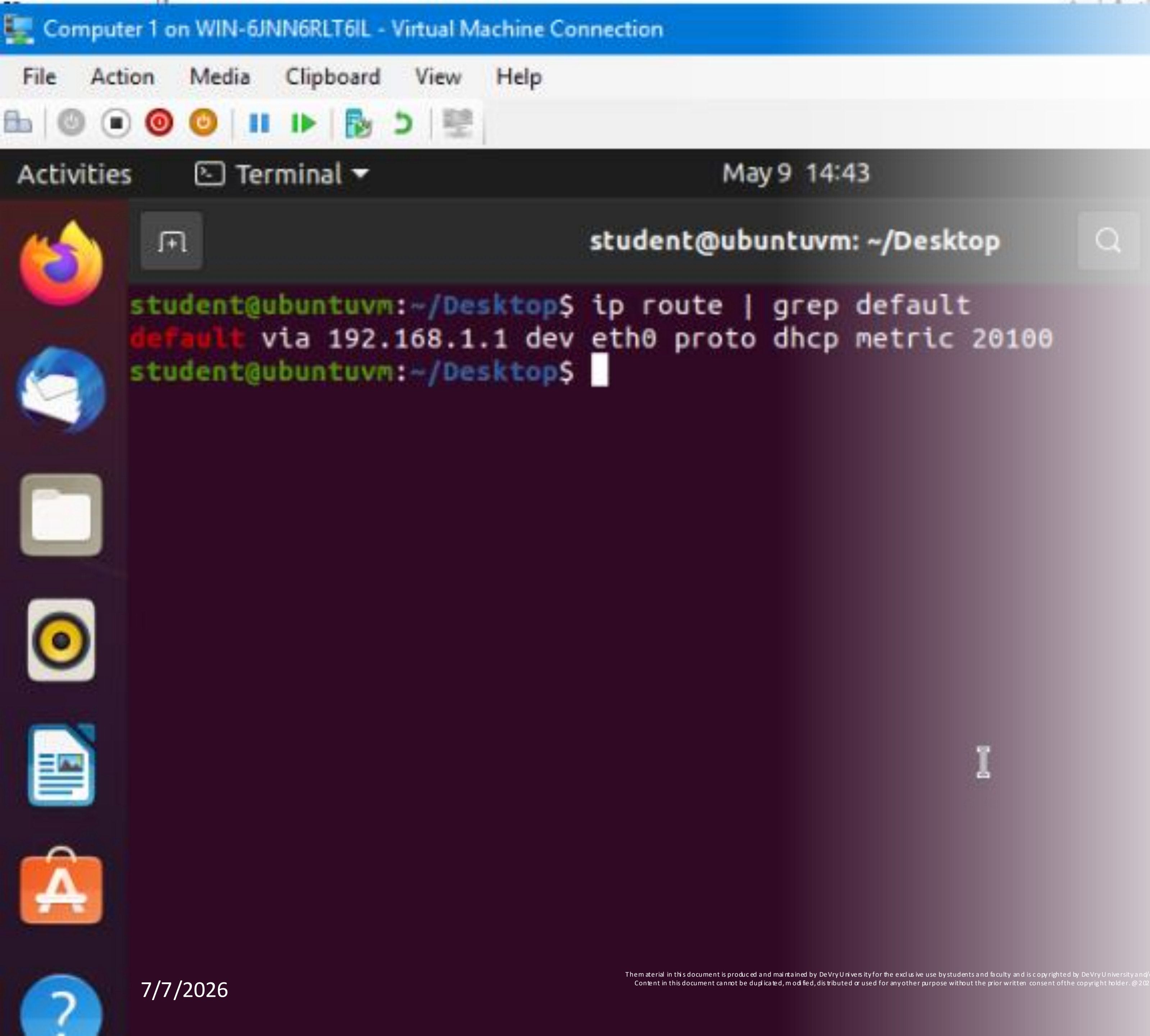
Introduction

This project is to implement an IPv4 addressing scheme to support a network. We will assign an IPv4 address to the SOHO Router virtual machine.

Summary

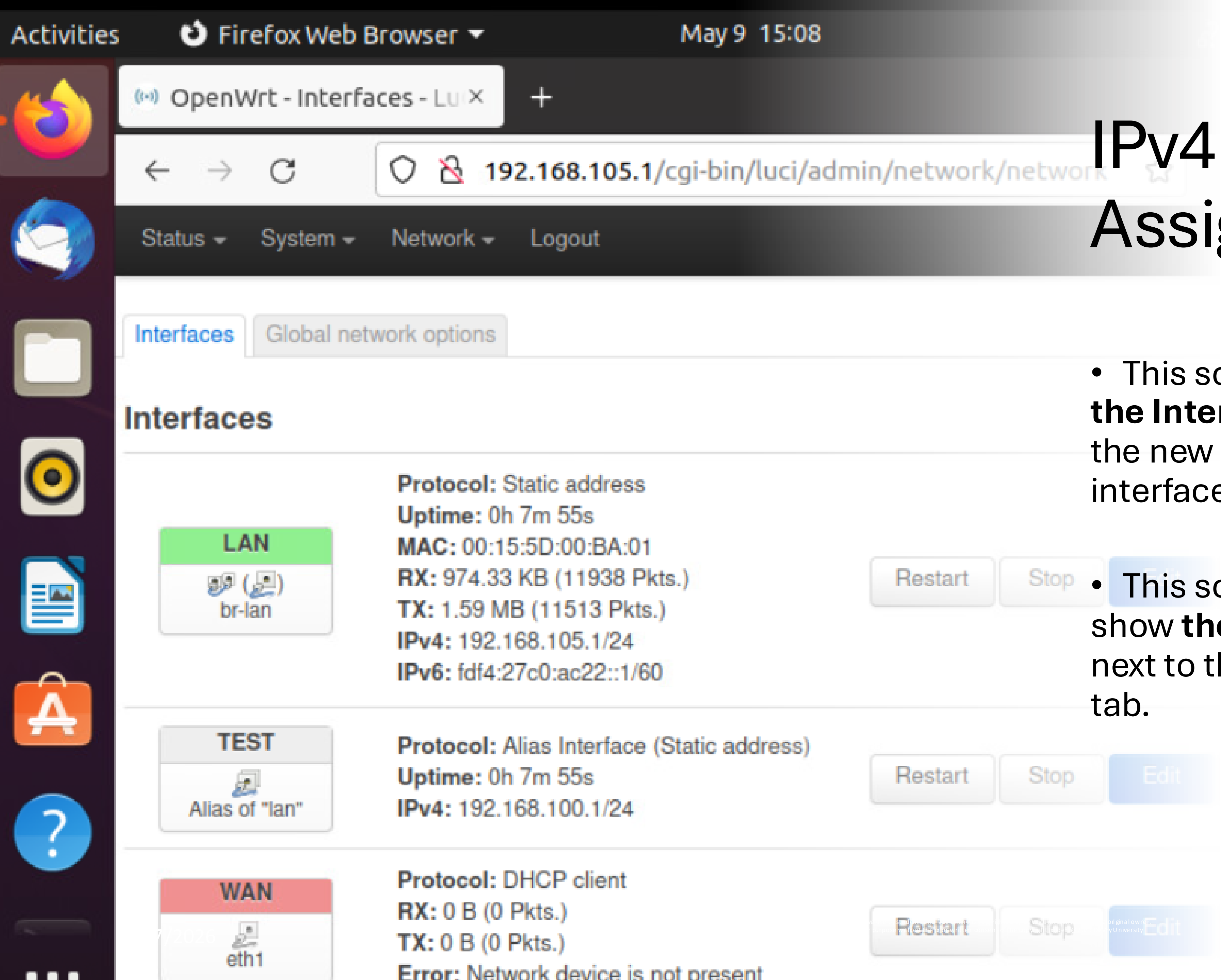
On Linux version, ubuntu computer 1. In the terminal we typed in the command *ip route | grep default*, this gives us the default IP address of the router we are interfacing with. Seen on screenshot Preparation. We then navigated to our Small Office/Home Office (SOHO) Router and changed the password for root by root. Now that we set a secure password, we navigated back to Computer 1, opened the computer's Web Browser, and typed in our IP address from our terminal to connect to our interface that is web-based for our router. We then applied changes to the Local Area Network (LAN), and now we have the desired IP address. Finally, we navigated back to our computer 1 terminal entered the *sudo ip link set eth0* down to lose connection, and then *sudo ip set eth0 up* to regain connection. The connection is complete, now we can navigate back to the Web browser to see our successful connection. Seen in the screenshot IPv4 Address Assignment.

Preparation



- This screenshot should include the terminal window that shows **the default gateway IP address**.

- This screenshot should also show **the date/time information** on top of the Terminal window.



IPv4 Address Assignment

- This screenshot should include **the Interfaces page** that shows the new IPv4 address on the LAN interface.
- This screenshot should also show **the date/time information** next to the Firefox Web Browser tab.

Conclusion

We learned that a SOHO router is the core network hardware device used to build a small home or office LAN. Devices connected to the SOHO router via a cable or wirelessly are typically configured as DHCP clients by default. They get an IP address and other network configuration parameters from the SOHO router's DHCP server. Creating challenges in locating SOHO addresses. They cannot be used by hosts. The first valid host address will be assigned to the LAN interface of the SOHO Router. Locating this was a challenge, getting to know what to look for is crucial so getting familiar with locating addresses was a great skill to gain. In all we learned to implement an IPv4 addressing scheme to support a network and assign an IPv4 address to the SOHO Router virtual machine.

NETW191 Course Project

Module 3

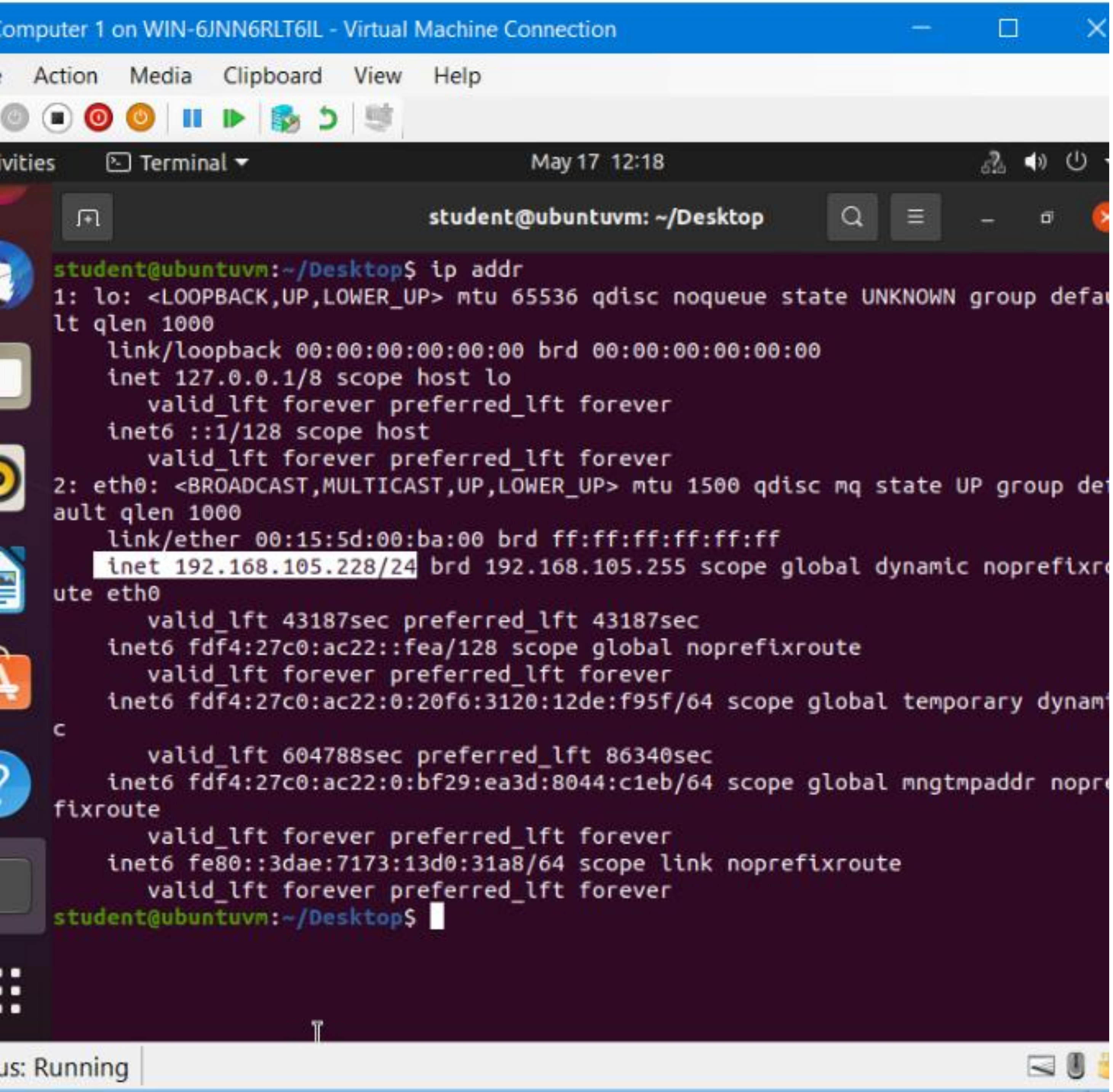
Connectivity Test

Introduction

This module aims to test the connectivity between the two computers and the SOHO Router.

Summary

A lot of functionality is built into a SOHO router. It contains a switch that supports multiple interfaces. It has routing functionality that connects your devices to the outside world (e.g., over a cable modem). It automatically performs network address translation between your LAN ports and the WAN port. Many SOHO routers include wireless access point functionality. The objective of this module is to test the connectivity between the two computers and the SOHO Router. First step, we typed *ip addr* in the terminal window to view the IP addresses of Computer 1 VM. Then we looked for the protocol family identifier inet when locating the VM's IPv4 address and found that it was 192.168.105.228/24, we see this in the first screenshot. The next step was to do the same on computer 2 and found that computer 2 IP address was 192.168.105.230/24. Now that Computer 1 VM, Computer 2 VM, and LAN port of the SOHO Router VM are on the same network. We used the ping utility program to verify the network connectivity among them. In the Terminal of Computer 1, we now verified connectivity to the SOHO Router VM (the default gateway) by using the following command *ping 192.168.105.1*. Same on Computer 2 Terminal with command *ping 192.168.105.228*. Verified connectivity was a success. Shown in screenshots 3 and 4.

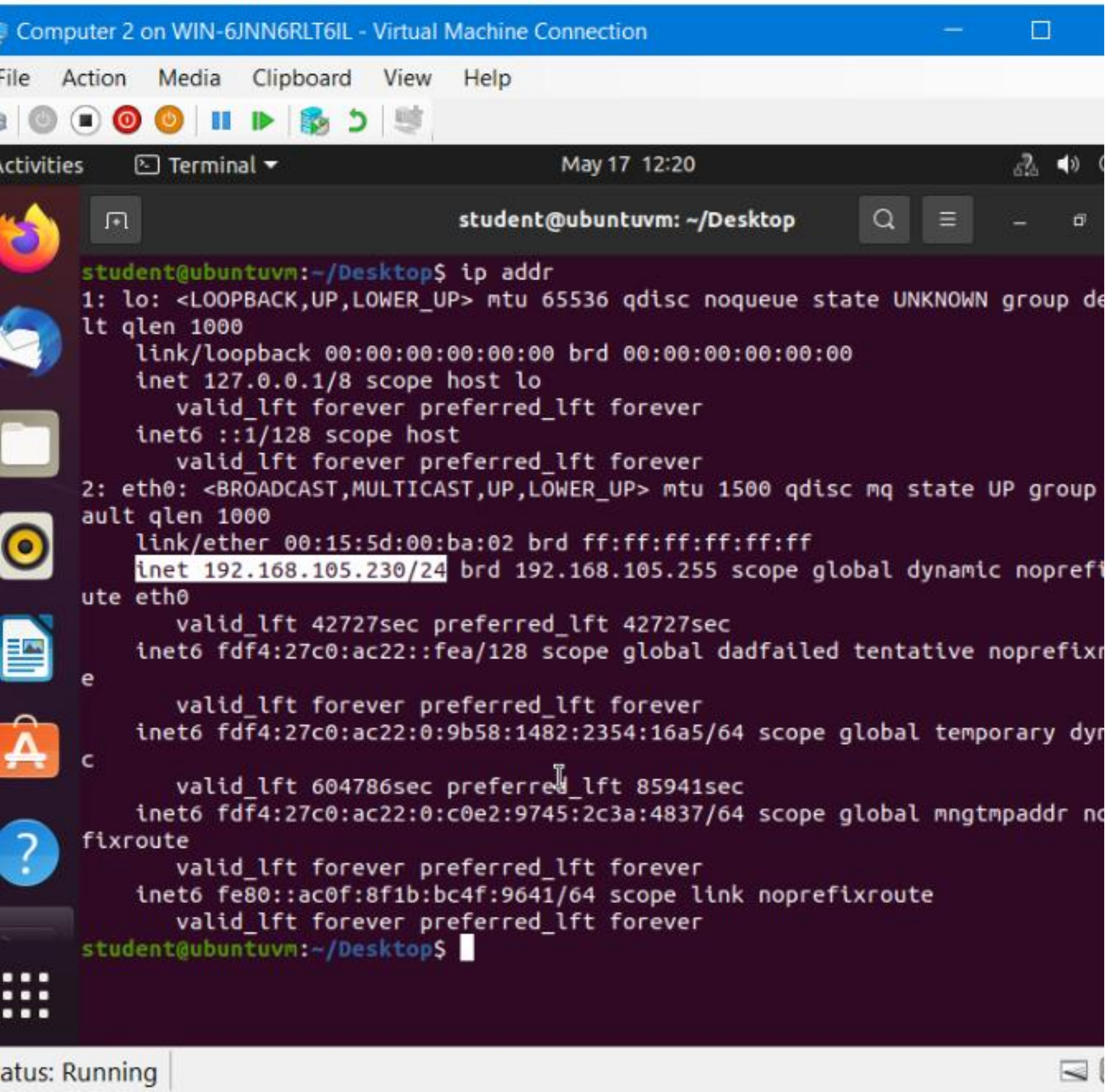


Dynamic IP Address Assignment

- Screenshot of **the Terminal window**.

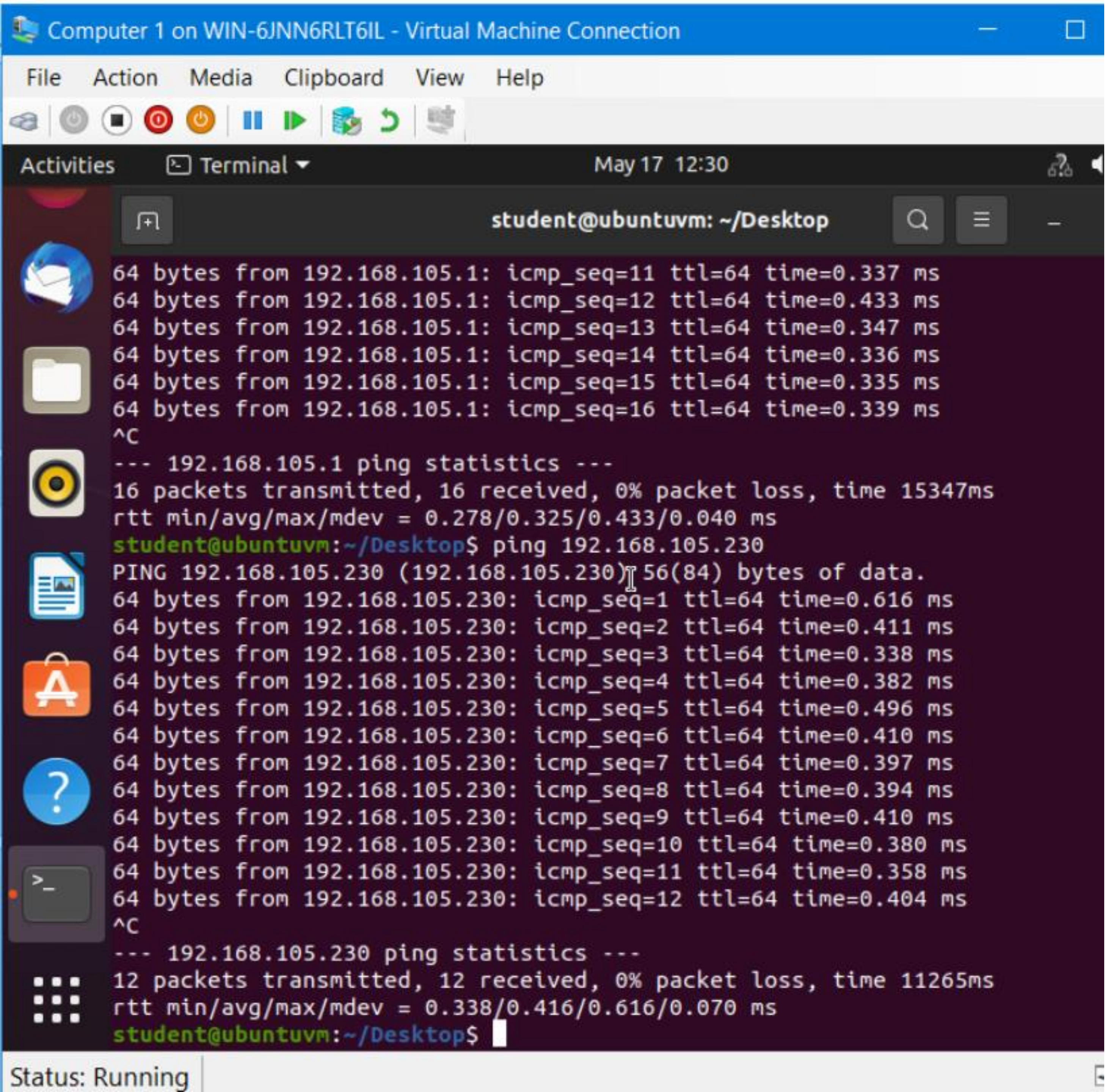
- This screenshot should show **the IPv4 address of the *<u>Computer 1</u>* VM**.

- This screenshot should also show **the date/time information** on top of the Terminal window.



Dynamic IP Address Assignment

- Screenshot of **the Terminal window**.
- This screenshot should show **the IPv4 address of the *<u>Computer 2</u>* VM**.
- This screenshot should also show **the date/time information** on top of the Terminal window.

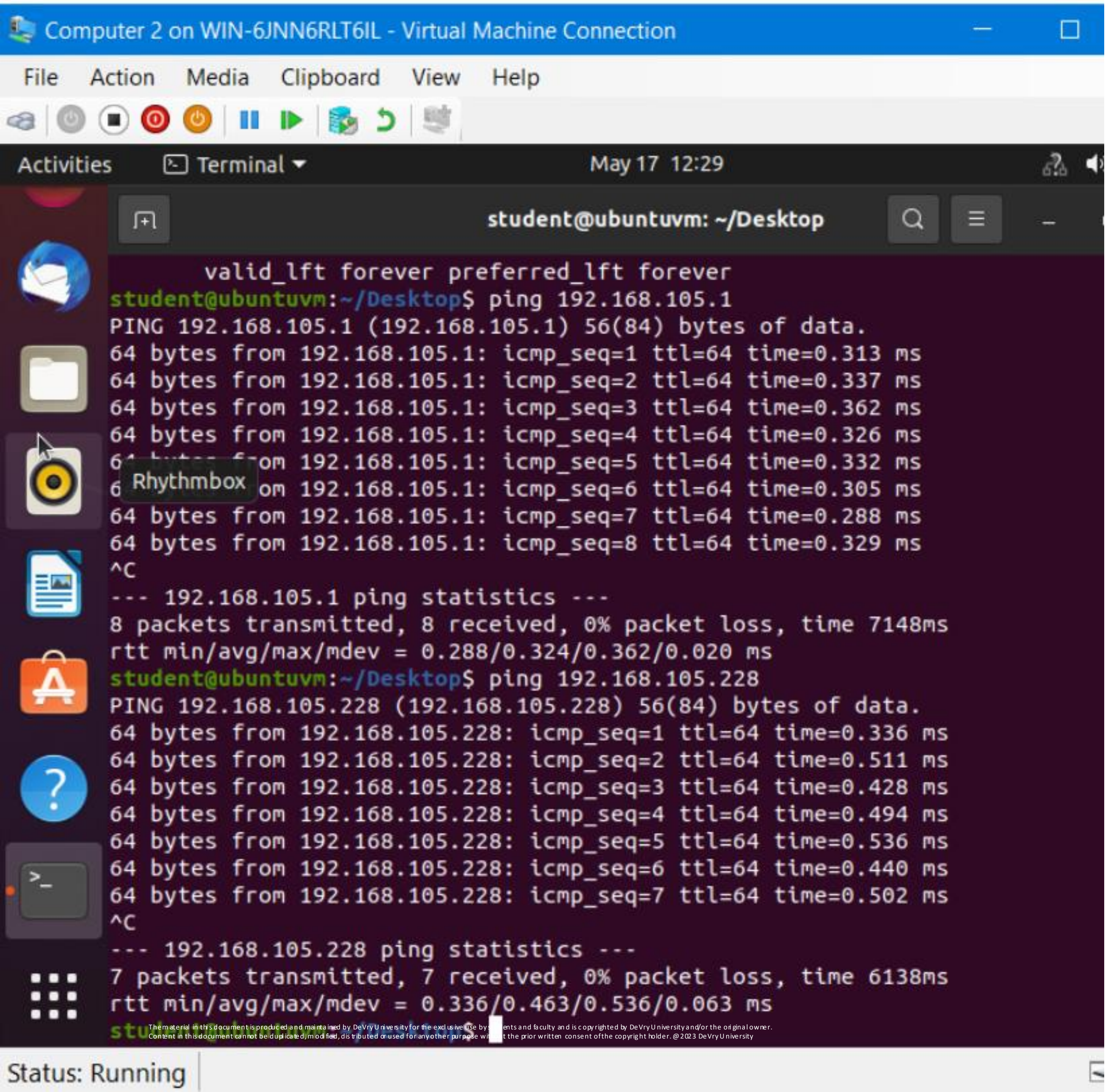


Connectivity Test

- Screenshot of **the Terminal window**.

- This screenshot should show 1) the connectivity test **FROM the *Computer 1* VM TO the *SOHO Router* VM**, and 2) the connectivity test **FROM the *Computer 1* VM TO the *Computer 2* VM**.

- This screenshot should also show **the date/time information** on top of the Terminal window.



Connectivity Test

- Screenshot of **the Terminal window**.
- This screenshot should show 1) the connectivity test **FROM the *Computer 2* VM TO the *SOHO Router* VM**, and 2) the connectivity test **FROM the *Computer 2* VM TO the *Computer 1* VM**.
- This screenshot should also show **the date/time information** on top of the Terminal window.

Conclusion

The objective of this module was to test the connectivity between the two computers and the SOHO Router. Challenges faced in this project, locating the IP address. Confusion with inet as a location for the IP address. A little confusion gave skills in implementing better performance and security.

NETW191 Course Project

Module 4

IP Subnetting and Loopback Interfaces

Introduction

In this module, we started with a class C network and divided its block of IP addresses into smaller blocks (i.e., subnets). Next, we configured two Loopback interfaces on the SOHO Router, using IP addresses of newly created subnets, to simulate two LAN segments.

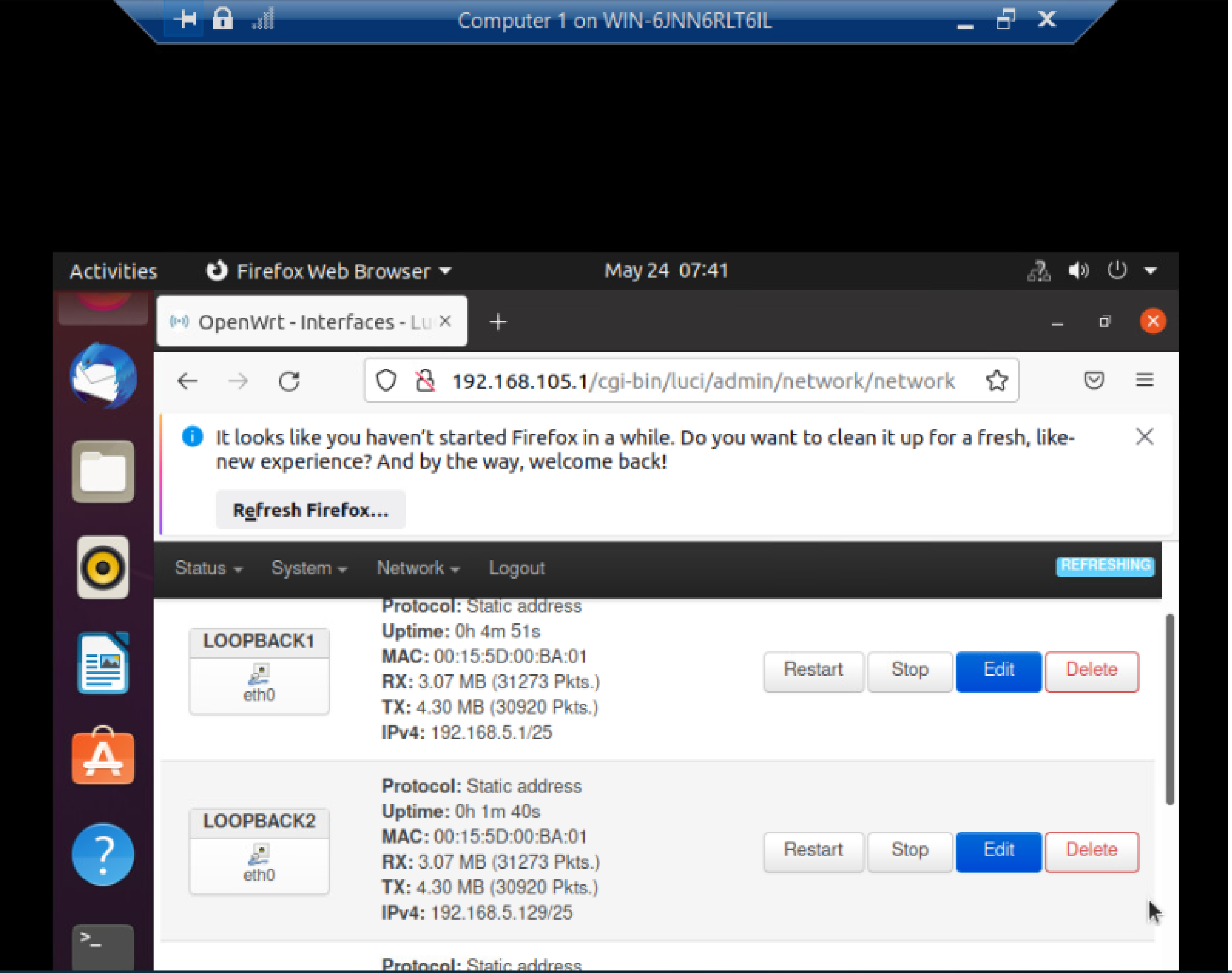
Summary

We started with a class C network and divided its block of IP addresses into smaller blocks (i.e., subnets). Next, we configured two Loopback interfaces on the SOHO Router, by using IP addresses of newly created subnets, to simulate two LAN segments. We were given an IP address block 192.168.5.0/24(network mask 255.255.255.0) and were asked to divide this block of addresses into two equal-sized subnets. This requires one bit to be borrowed from the host portion. When the borrowed (subnet) bit is 0, it represents the first subnet. When the borrowed (subnet) bit is 1, it represents the second subnet. With the borrowed bit being added to the network portion, both subnets have a prefix /25 or network mask 255.255.255.128. We already know /25 is 255.255.255.128. Due to changing our net mask into binary numbers. 11111111.11111111.11111111.10000000. Now that we have our network mask and our network addresses, we will be able to plug in the broadcast addresses and the first and last usable hosts. Now that we have all our information, we established a connection and tested out ping in the terminal shown in screenshots 1 and 2.

IP Subnetting

- This table should include **two /25 subnets**, listing
  - Subnet notation
  - Network address
  - First usable host address
  - Last usable host address
  - Broadcast address

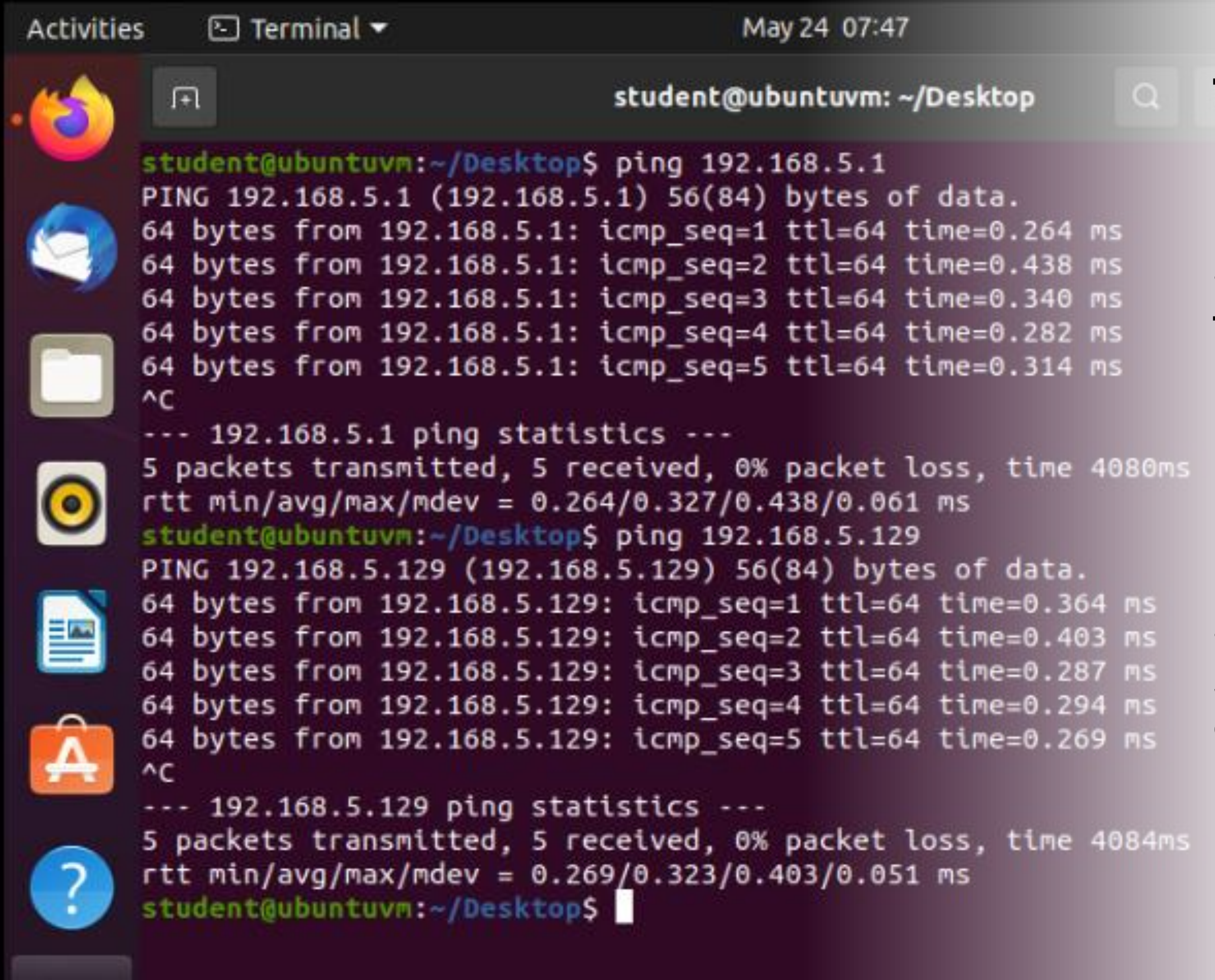
Subnet ID	Network Mask (/prefix)	Network Mask (Dotted decimal)	Network Address	First Usable Host Address	Last Usable Host Address	Broadcast Address
0	/25	**255.255.255.128**	192.168.5.0	192.168.5.1	**192.168.5.126**	192.168.5.127
1	**/25**	255.255.255.128	**192.168.5.128**	192.168.5.129	192.168.5.254	**192.168.5.255**



Loopback Interfaces

- This screenshot should show **both Loopback1 and Loopback2 interfaces** and their correct **IPv4 addresses**.
- This screenshot should also show **the date/time information** on top of the desktop window.

Connectivity Tests



- This screenshot should show **two successful ping tests**:
  - **from** the Computer 1 VM **to** the Loopback 1 interface
  - **from** the Computer 1 VM **to** the Loopback 2 interface

- This screenshot should also show **the date/time information** on top of the desktop window.

Conclusion

We started with a class C network and divided its block of IP addresses into smaller blocks (i.e., subnets). Next, we configured two Loopback interfaces on the SOHO Router, by using IP addresses of newly created subnets, to simulate two LAN segments. Challenges I faced during this project were dividing the block of addresses into two equal-sized subnets. Not knowing we borrowed a bit to do so, this strengthened my skill in subnetting, organization with subnetting tables, and with connectivity testing.

NETW191 Course Project

Module 5

Visio Network Diagram

Introduction

In this module, we used Microsoft Visio to create a network diagram that depicts the interconnection of the Computer 1 VM, the Computer 2 VM, and the SOHO Router VM.

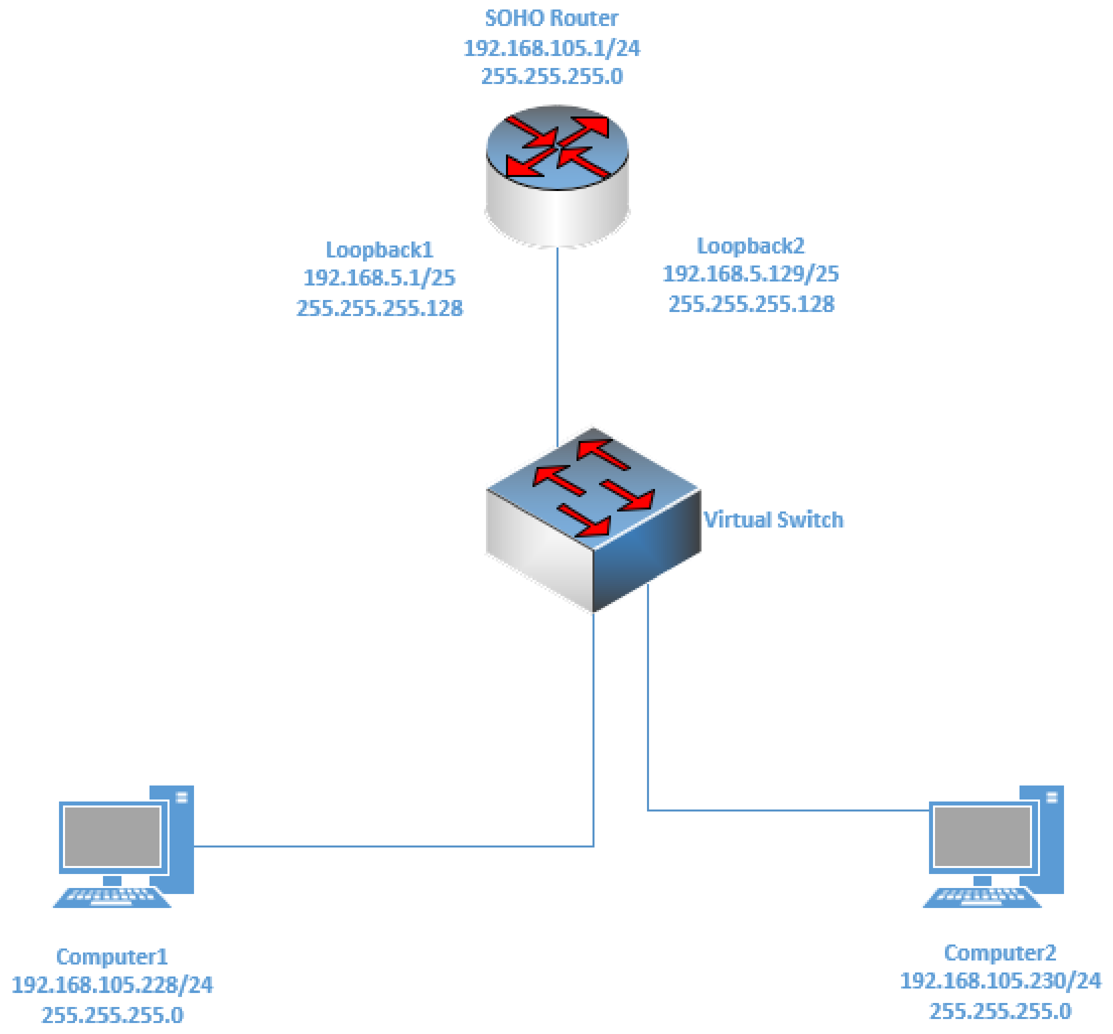
Summary

In Course Project Module 3, we performed a connectivity test between two computers and an SOHO Router and in Course Project Module 4, we configured two Loopback interfaces on the SOHO Router, by using IP addresses of newly created subnets, to simulate two LAN segments. In this module, we created a network diagram that depicts the interconnection of the Computer 1 VM, the Computer 2 VM, and the SOHO Router VM using Microsoft Visio. Microsoft Visio is a diagramming and vector graphics application that lets you transform hard-to-understand text and tables into organized visual diagrams. In our Visio Network Diagram, we created a diagram using everything we have been working on throughout our modules, creating a well-organized and understandable visual diagram of our network.

Microsoft Visio Network Diagram

- This diagram should illustrate **the interconnection** of the Computer 1 VM, the Computer 2 VM, and the SOHO Router VM along with proper **IP addresses** and **labeling**.

- This diagram should also show **your initials** in the bottom right corner.



Conclusion

In our Visio Network Diagram, we created a diagram using everything we have been working on throughout our modules, creating a well-organized and understandable visual of our network. During this project, a challenge I faced, locating different types of wiring. Learning to provide a comprehensive view of data so others would know what I was creating was a challenge. Using Visio created new skills in Microsoft. Having the ability to communicate ideas through visual diagrams sets skills in quickly identifying problems to quickly developing solutions. In a work environment improving organizational efficiency is crucial and is a great skill to have in cybersecurity.

NETW191 Course Project

Module 6

SOHO Wireless Network Security

Introduction

In this module, we explore some router configurations that could be used to add protection to a SOHO wireless network.

Summary

In this project module, we explored router configurations that could be used to add protection to a SOHO wireless network. We visited https://www.tp-link.com/us/support/emulators. The emulator is a virtual web GUI where you can experience the TP-Link product management. A great place to learn how to secure networks. We first selected our hardware and were shown the default username and password. A security concern, we are shown how vulnerable defaults are, as they can be searched for by anyone through websites like https://www.routerpasswords.com/. We then navigated to DHCP Settings, to the client's list where we were given the choice to select how we would configure our IP addresses. Either through configuring IP addresses manually (static IP) or using a DHCP server (dynamic IP). Now in device management, we navigated to the Wireless MAC Filter, showing MAC addresses that were being filtered by either having the administrator allow access to deny access to the network. Right back to the left navigation pane, we navigated to Wireless 2.4 GHz, Wireless Security. When given other choices for network security we are strongly encouraged to select WPA2-PSK AES (Advanced Encryption Standard) encryption as AES unites with multiple rounds of encryption making the network not worth hacking.

SOHO Wireless Network Security 1 of 3

1. What are the factory default username and password of a TP-Link router? Why is it important to change the default username and password of a SOHO router?

Answer: The default username is **admin** and the password is **admin**. Changing the default username and password is important because your router's default information is public knowledge. Anyone can find the default username and password to a router on a website like TP-Link or ROUTER Passwords.com.

2. To protect a SOHO wireless network with a small number of devices, which address management method provides more control, configuring the device IP addresses manually (static IP) or using a DHCP server (dynamic IP)? Why?

Answer: The method that provides more control in protecting your SOHO wireless network is to configure the device IP addresses manually (Static IP). With Static IP, Network admins manually configure IP addresses making them more secure, DHCP can be difficult to track due to the IP addresses being assigned by default.

SOHO Wireless Network Security 2 of 3

3. What does MAC filtering do? If needed, when would you use deny filtering rules, and when would you use allow filtering rules? What happens to devices that want to connect, if the “Allow the stations specified by any enabled entries in the list to access” function is enabled but there are no entries in the list?

Answer: MAC filtering takes the MAC addresses connected to a network, your laptop, computers, smartphones, routers, etc. For anything that connects to your network, MAC filtering will pick up on the MAC address, and depending on what your table addresses are assigned. The MAC filtering will either allow access or deny access to the network, addresses can be blocked as well. If needed when letting a device on the network is allowed by the administrator, but the administrator does not want to allow access anymore, the administrator can apply MAC filtering the deny filtering rule to deny access. If there are no entries to the list, then no devices will be able to connect. Since there are no entries including your computer and the function is enabled, you would have to manually connect via ethernet cable and include your MAC address to the filtering list or turn the feature off temporarily so you can add addresses.

4. What wireless security settings are displayed on the Wireless Security page? Which one is recommended by the vendor? Why?

Answer: On the wireless security page, it is clearly stated “For network security, it is strongly recommended to enable wireless security and select WPA2-PSK AES (Advanced Encryption Standard) encryption. AES unites with multiple rounds of encryption making the network not worth hacking for it will take way too long.

SOHO Wireless Network Security 3 of 3

5. Among the configurations you explored in this module, which one is a true security function? Why?

Answer: The true security function I would argue would be the WPA2-PSK AES, due to it incorporating multiple rounds of encryption for outsiders trying to get in. Encryption protects data, making it unreadable to unauthorized people and entities, scrambling and protecting our information, I would argue a true security function.

6. What would you do to protect your wireless network at home? Why?

Answer: What I would do to protect my wireless network at home would be to keep updating my router firmware and create a guest network for my children or visitors for protection from vulnerable websites and malware.

Conclusion

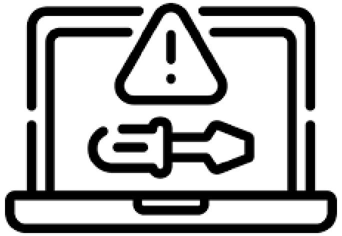
In this project module, we explored router configurations that are used to add protection to a SOHO wireless network. All these techniques can be used to make a SOHO network more secure. Changing the default username and password so that malicious users can not access your network is a huge step everyone should take. A challenge that I faced in this module was the reality of not knowing how important securing our network is and the steps to do so. Starting in my home with my router was my first step in securing my network. Routing configurations helped me gain skills in filtering and setting conditions in a network. These qualities are essential due to networks having large open ports for security threats, this skill would help companies tremendously.

Challenges/Lessons Learned

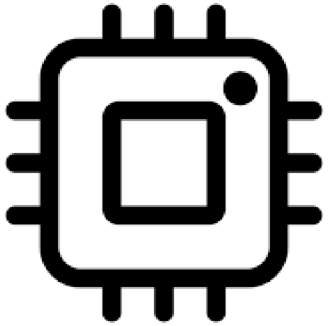
A challenge I faced during this project was dividing the block of addresses into two equal-sized subnets. How I overcame this challenge was I looking at various IP subnets and kept practicing until I felt confident. A lesson I learned during this project is practice makes perfect.

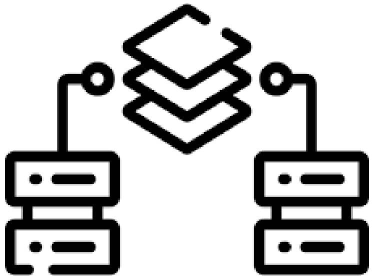
Career Skills Gained

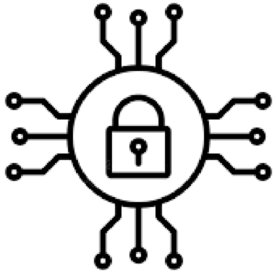

Networking


Troubleshooting


Cloud Computing


Hardware


Virtualization


Network Security

Conclusion

Progress in Networking has been rapid in recent years, and do not expect it to slow down anytime soon. This foundation of emerging technologies, including IoT has enormous future potential. As the workplace becomes digitalized, success is fueled by the ability to use technology to make data-driven decisions. This makes having skills in Information Technology and Networking essential to everyone. This course project covers the fundamentals of networking that play a vital role in the IoT world. Building a network's design and development process includes SOHO router configurations, subnetting, connectivity testing, networking documentation, and SOHO wireless network security.

References

https://www.tp-link.com/us/support/emulator/

https://www.routerpasswords.com/

Rasmussen, Anders Ingeman, A. (2010). *Microsoft Visio 2010 – Open source Visio alternatives.* *https://www.osalt.com/visio*